

What EDA Reveal about Flickr

Through Communities, Timing, and Metadata

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- DATASET OVERVIEW -

DATA SOURCE & FORMAT

XML

Group_57.xml → Photo records
(metadata fields)

+

JSON

Group_57.json → Geolocation and tags



merged data using **Post_ID**

32907

records

18

attributes

96

duplicate
Post_ID

ATTRIBUTES

Identifiers

Post_ID, User_ID, Secret

Network

Server, Farm

Temporal

Post_Date, Taken_Date, Min_Taken_Date

Location

City, Country, Longitude, Latitude

Content

Title, Description, Tags

Access

Is_Public, Is_Friend, Is_Family

FLICKR REQUIRES ACCURATE AND RELIABLE INFORMATION BEFORE IT CAN ENHANCE PERSONALISATION.

DATA QUALITY AND INTEGRITY

Insight 1

'Posts may appear older than the actual photos due to timestamp validation gap'

Insight 2

'Users' location is indicated by the City and Country fields, not the location of the photos'

Insight 3

'Three fundamental user categories are revealed by metadata completeness and each requires a unique product reponse'

Findings

- **325 posts** have Post_Date < Taken_Date
- **3,601 days** negative gap
- **City/Country** → home location, **not** photo location
- **Missing** → **44%** City & **41%** Country

Reasons

- Influence data-based search → **inaccurate** timestamps
- Misleading location text → **harm** location-based discovery
- Must be carefully evaluated → some fields are **unreliable** to use directly

Recommendation

- **Eliminate** timestamp edits
- **Prioritize** GPS as ground-based source for location
- **Clarify** between the user's home and photo location

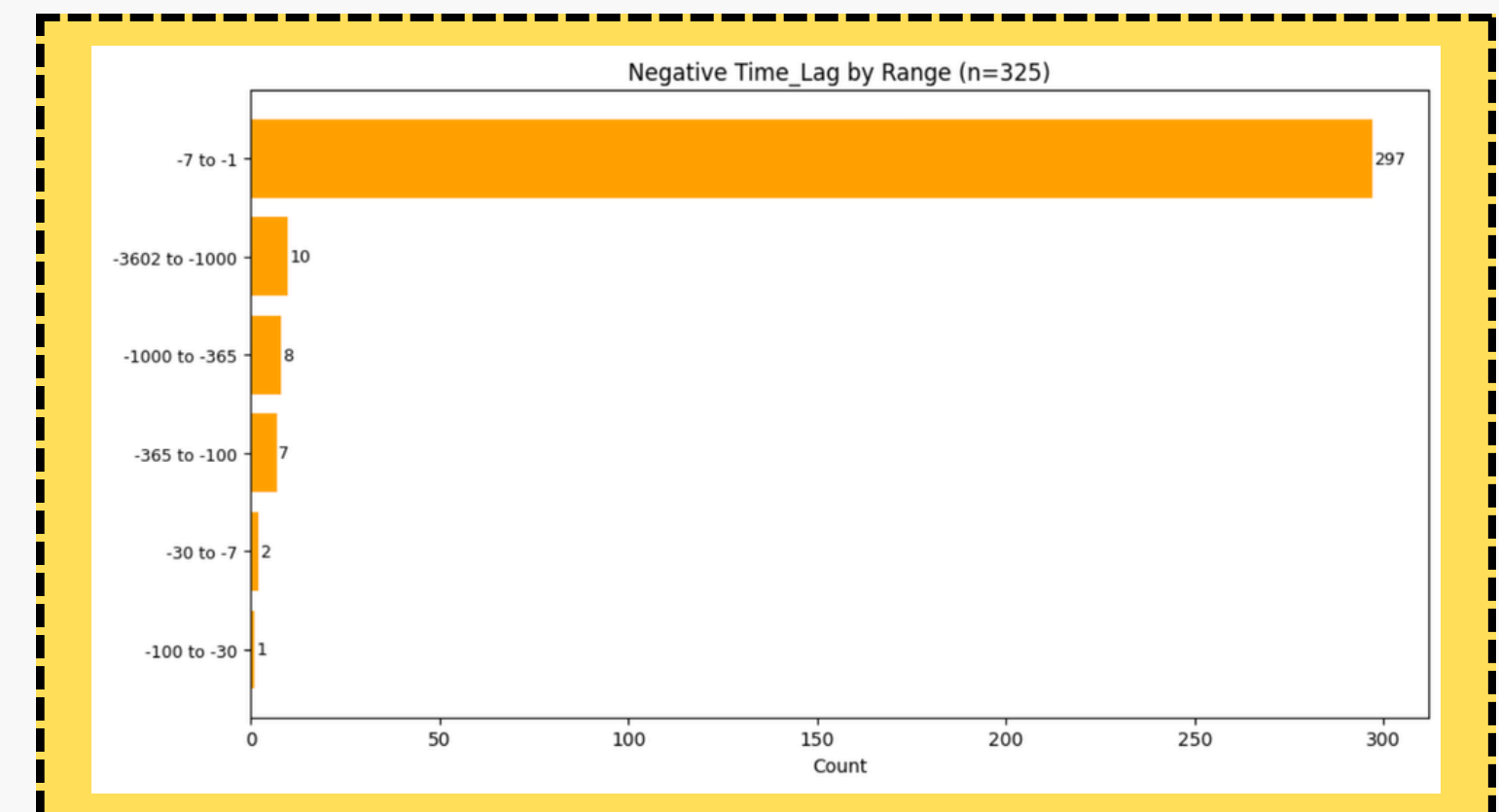


Figure 1. The Negative Time_Lag Histogram

WHAT USERS PHOTOGRAPH MATTERS MORE THAN WHERE THEY LIVE.

UPLOAD TIMING FOR USER CONTENT BEHAVIOUR

INSIGHT 7

Posting urgency is driven by content type, instead of geography where the photo was taken

- Travel and event photos get uploaded within 15 days
- Nature and wildlife photos sit for 290 to 304 days before going up
- This gap looks exactly the same in the UK, Netherlands, and US

BUSINESS VALUE

- Notification timing should follow content type, not location
- A wildlife photographer and an event photographer each need reminders that match how they actually shoot

FINDINGS

- A UK photographer posting Scotland trip photos uploads fast (same person), posting local park photos takes months
- Geography explains nothing here, content does
- Country-based notification logic is less accurate

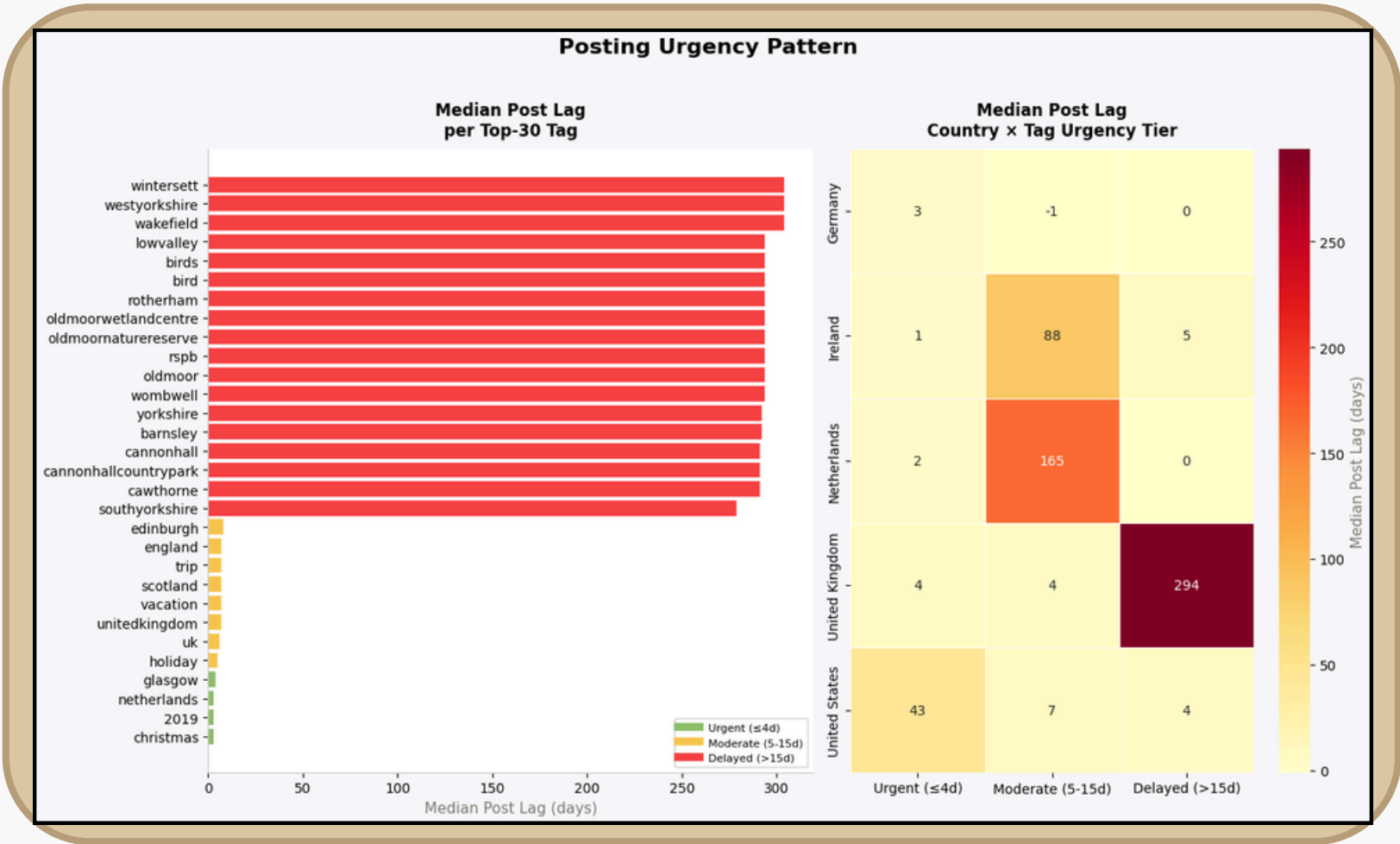


Figure 2. The Posting Urgency Pattern

FLICKR OUGHT TO DISCONTINUE ACTING ASSUMING THAT ALL USERS BEHAVE THE SAME WAY.

USER ARCHETYPES & BEHAVIOR

Insight 3

Three fundamental user categories are revealed by metadata completeness and each requires a unique product response

Insight 8

Burst uploader constitute a valuable archetype that Flickr is now unable to differentiate from bots and runs the risk of completely alienating users

Findings

- Title, tags, & description → filled in by serious photographers
- Tags & description → frequently skipped by casual uploaders
- Bot-like or automated accounts → high repeated metadata patterns

Reasons

- Single experience → not suited for every user
- High-value users → risk of viewing them as abnormal
- Different product reactions → driven by user behavior

Machine Learning 1 User Archetype Classification

Can users be categorized by Flickr into:

- bot/scripted accounts,
- casual uploaders, and
- serious photographers

utilizing upload behavior and metadata completeness?

- Customize encouragement and onboarding processes
- Provide greater support
- Detect suspicious account

Business Value

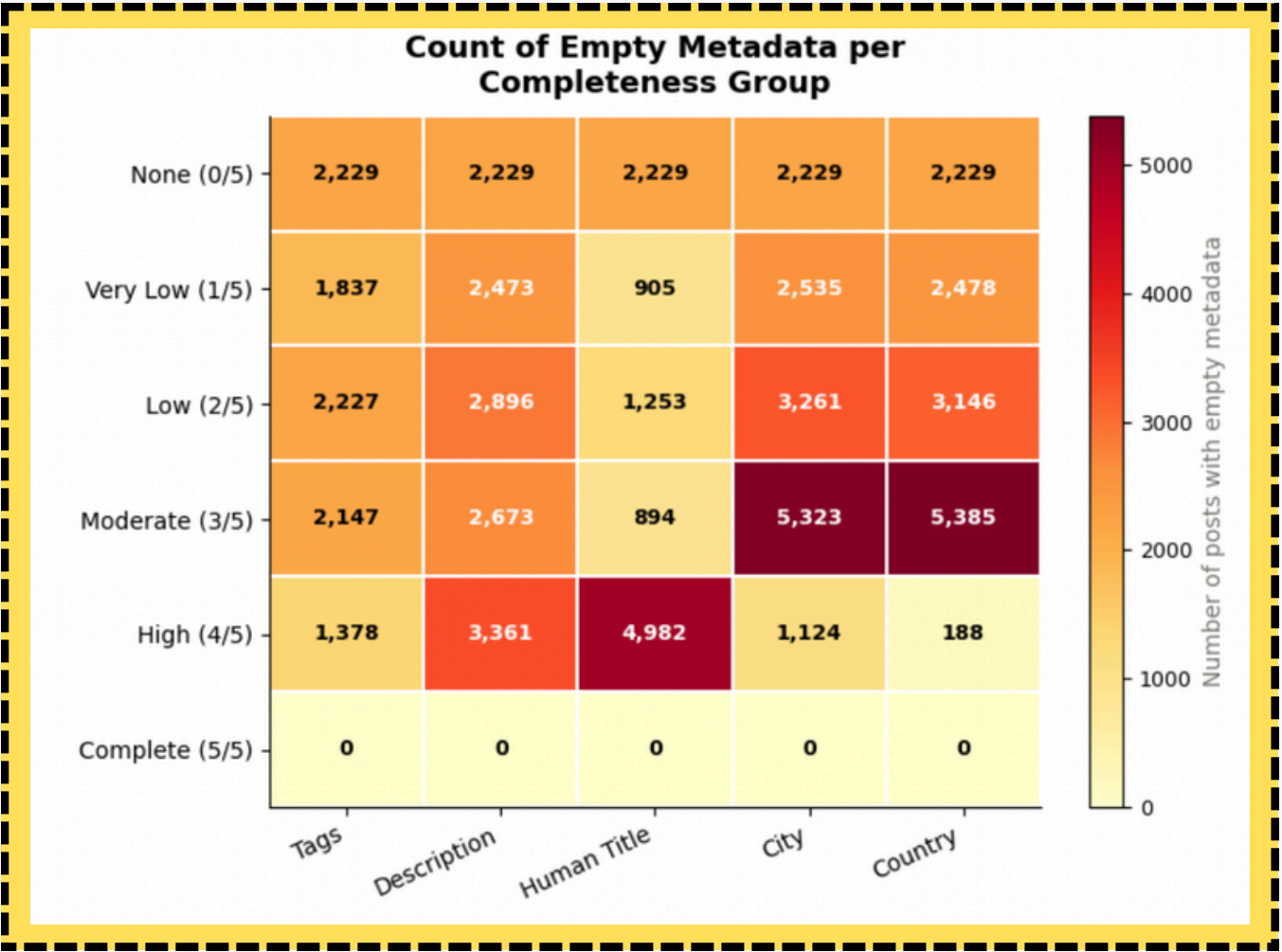


Figure 3. The Metadata Completeness Chart

THE POSTS THAT MATTER MOST IN THE MOMENT ARE OFTEN THE HARDEST TO FIND.

METADATA QUALITY & SEARCH

Insight 4: 10.7% of posts carry meaningless Title

- Repeated title such as "rcl-acv" (x169) give search engines nothing to work with
- Search ranks by newest instead of most relevant
- Affects every single user's search experience

Insight 5: Users fast posting have the worst metadata

- Users who upload the fastest fill in the **least** metadata such as Title, Description, and Tags.
- It shows the user efforts to fill out details when they need to upload the photos in hurry.

Findings

- 10.7% of all titles are auto-generated filenames like img_, dsc_, photo_ with zero descriptive value
- The fastest-uploading clusters have the worst metadata scores across the board
- Photos already have a URL in the dataset, making computer vision technically feasible right now

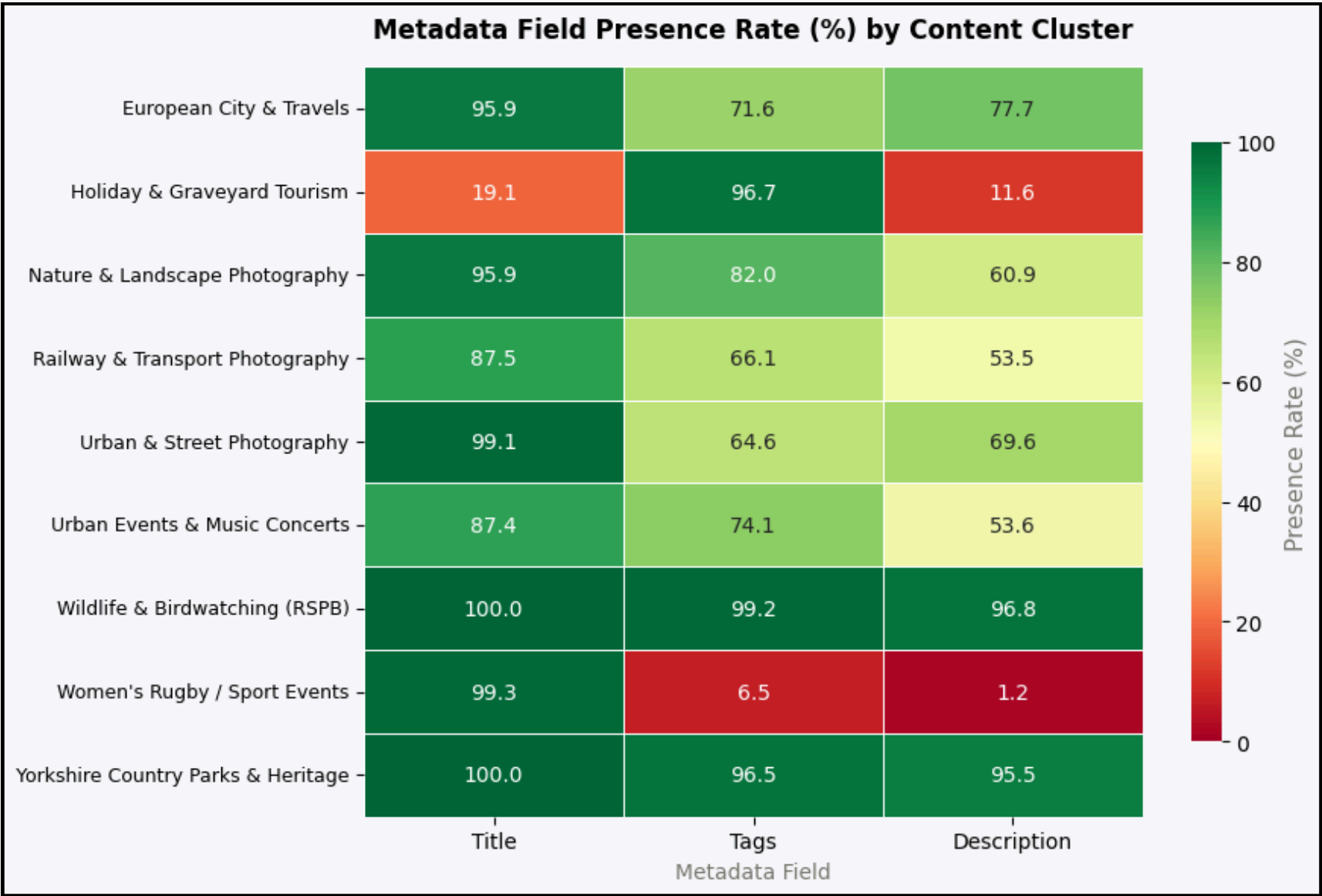


Figure 4. The Metadata Presence Rate Chart

THE POSTS THAT MATTER MOST IN THE MOMENT ARE OFTEN THE HARDEST TO FIND.

METADATA QUALITY & SEARCH

Machine Learning 3: Automated Metadata Enrichment via Computer Vision

Can we automatically generate tags, titles, and descriptions for posts with missing or poor metadata, using the photo itself as the source?

Business Value

- Thousands of posts that are currently invisible in search become discoverable
- AI reads the photo directly, so suggestions are based on what's actually in the image
- The fix happens at upload moment, exactly when the user is least likely to fill things in manually

USERS REFRAIN FROM POSTING WHEN THEY SHOOT, WHICH AFFECT WHEN FLICKR SHOULD GET IN TOUCH WITH THEM.

TEMPORAL PATTERN & NOTIFICATIONS

Insight 6

Flickr’s peak engagement hour is from 8 to 10 pm on Saturday and Sunday considering users take photos on weekend afternoons and post at night

Insight 8

Burst uploader constitute a valuable archetype that Flickr is now unable to differentiate from bots and runs the risk of completely alienating users

Business Value

- Increase the rates of opennes and action
- Decreases notifications taht are not timely
- Encourages individualized communication

Findings

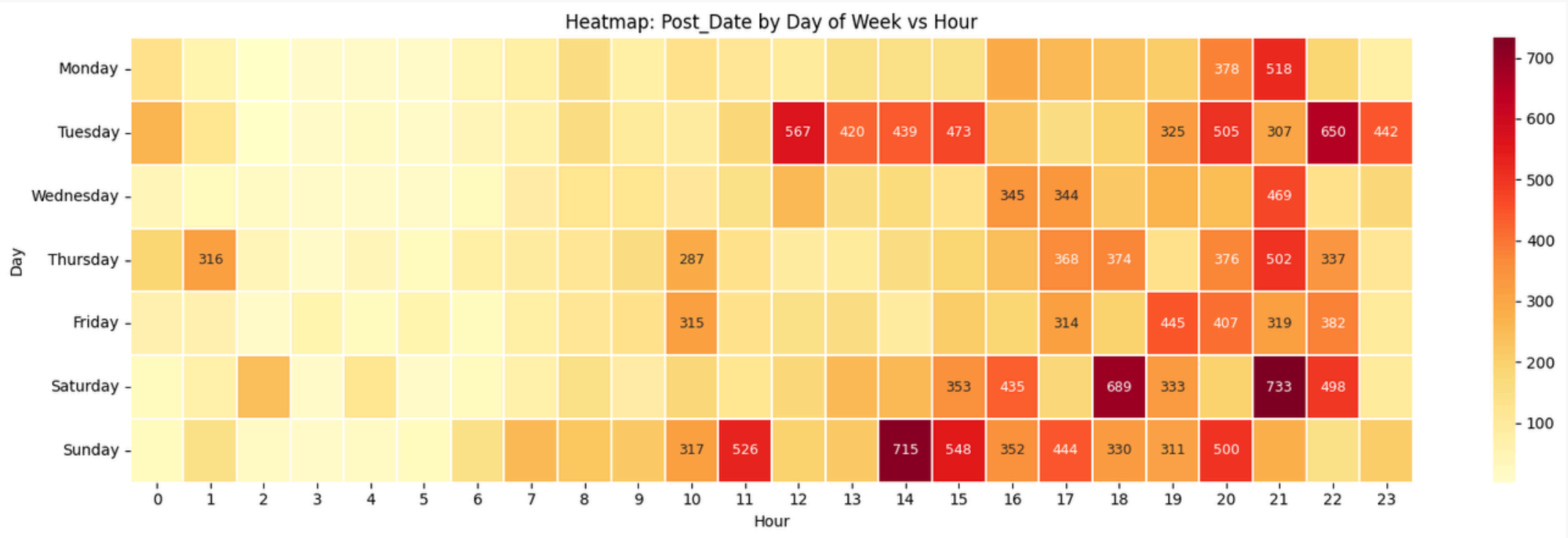
- Photo capture peaks → Sunday afternoon
- Highest uploads → Saturday & Sunday nights
- Substitute catch-up opportunity → Tuesday evening
- Burst uploaders → different timing than ordinary users

Reason

- Ineffective notification schedule
- User timing behavior corresponds to predictable patterns

Machine Learning 5
Optimal Notification
Time Prediction

Can Flickr utilize posting behavior to determine each user’s optimal notification or email timing window?



CONNECT THE USERS WITH THE SAME INTEREST

CONTENT COMMUNITIES

Machine Learning 2: Community Detection & Group Recommendation

Can we use what users already post to group them into communities, then point them to Flickr groups they'd actually want to join?

Insight 9: The data groups itself into 9 same interest communities

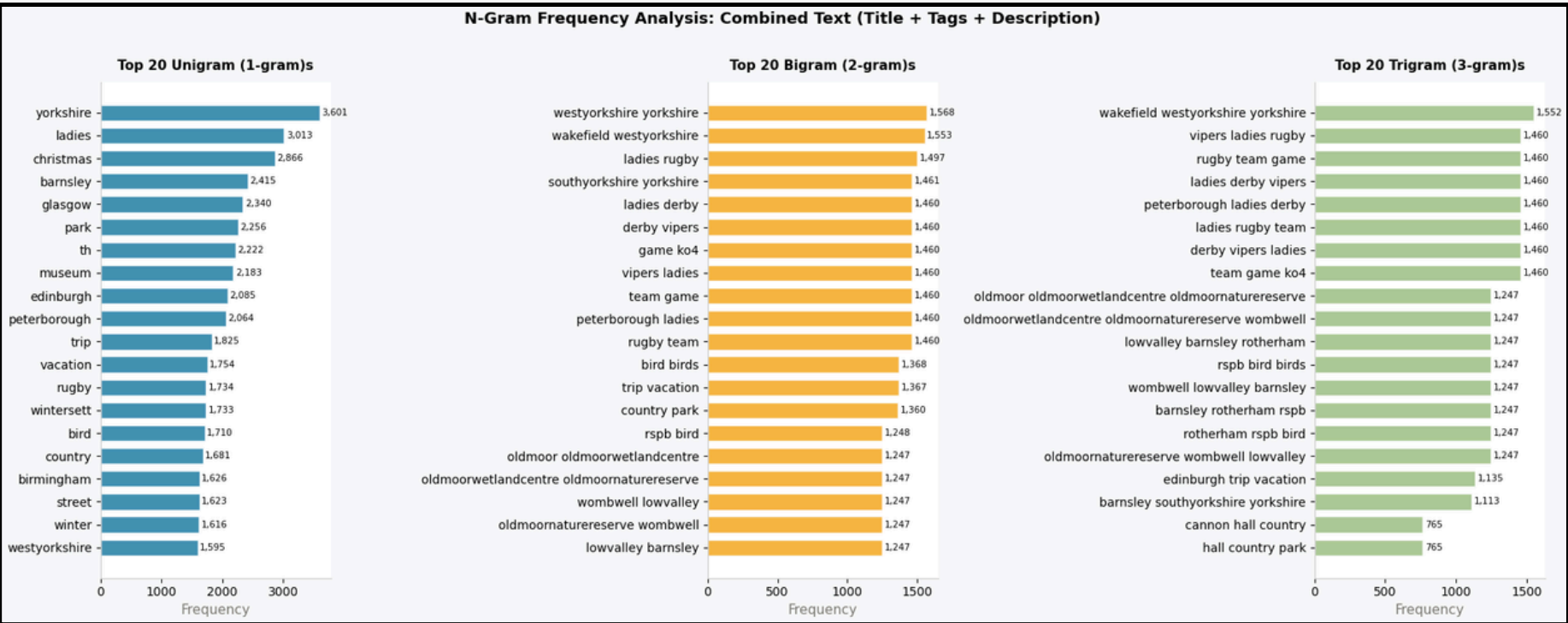
By analysing only titles, tags, and descriptions, the data naturally splits into 9 distinct communities. Users in smaller communities post more frequently and write more detailed metadata

Business Value

- A prompt like "47 people are sharing similar photos this week" gets users to discover groups on their own
- For smaller communities, a small but active group is enough to make users feel like they belong and that's what makes them stay
- This runs fully on data that's already there

Findings

- 9 real communities found just from post text, no extra input from users needed
- Users in niche communities are the most active and detailed on the platform
- Right now, nothing on Flickr connects these users to each other



THE MOST VALUEABLE USERS CAN BE IDENTIFIED BEFORE THEY HIT FRICTION

USER VALUE & MONETISATION

Insight 3

Three fundamental user categories are revealed by metadata completeness and each requires a unique product reponse

Insight 8

Burst uploader constitute a valuable archetype that Flickr is now unable to differentiate from bots and runs the risk of completely alienating users

Insight 10

Targeting niche interest cluster is the most effective retention approach available since they provide disproportionately high-quality content

Findings

- Improved metadata and engagement patterns displayed by serious contributors
- Burst uploaders not only be troublesome but also valuable
- Niche clusters created the best content

Business Value

- Facilitates speedy pro upselling
- Focuses on users at periods of high engagement
- Directly connects behavior to monetization strategy

Reasons

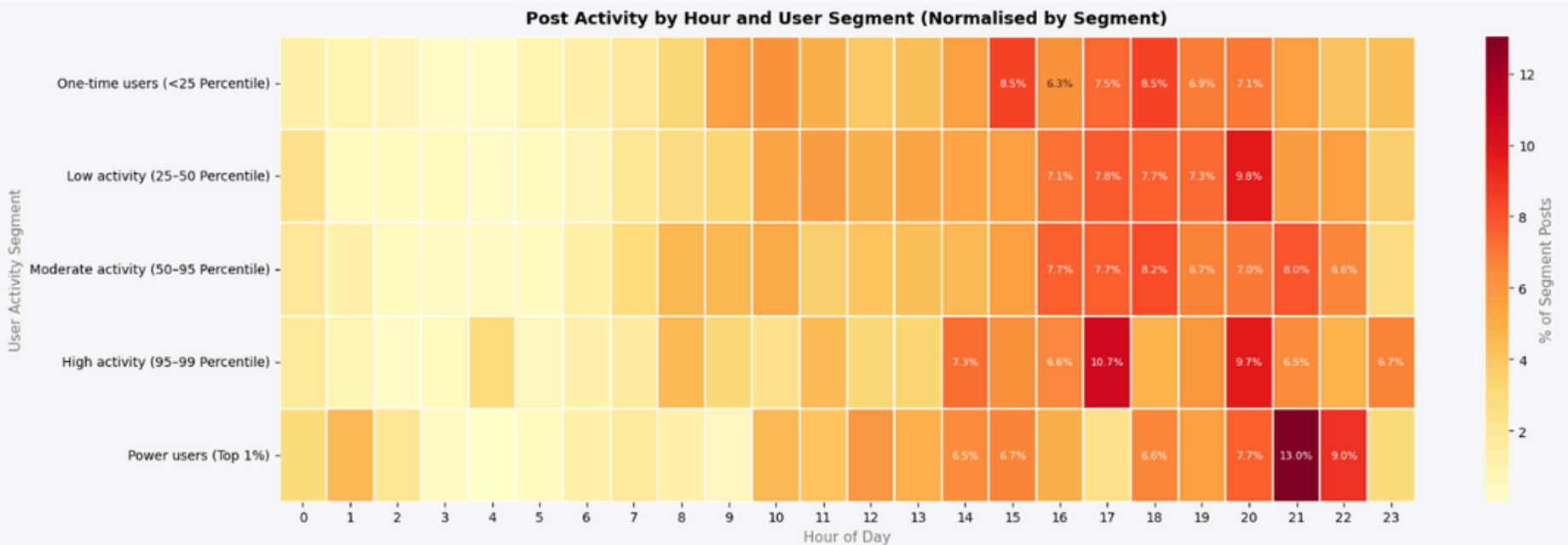
- Ability to recognize valuable users before they become irritated
- Possibility to target retention and monetization effectively

Machine Learning

4

Pro Upgrade
Candidate
Prediction

- Can Flickr identify users who are likely to exceed free limits and become strong Pro upgrade candidates?



**"THE GOAL IS TO TURN DATA INTO
INFORMATION, AND INFORMATION INTO
INSIGHT."**

— Carly Fiorina, ex CEO HP